Stelios Zacharioudakis

MACHINE LEARNING RESEARCH & ENGINEERING

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SUMMARY

Machine learning engineer with a BSc in Computer Science and pro bono software engineering experience delivering a member-services platform. Build PyTorch research tools, reproducible generative-model evaluations, and full-stack applications using Python, React, Node.js/Express, PostgreSQL, and Linux deployment. Independent research explores latent world models.

WORK EXPERIENCE

Paphos Medical Association | ***Head Engineer (Pro Bono)*** **Jun 2022 - Jul 2026**

- Delivered and donated AsklepiosMed, a web platform that provided the Paphos Medical Association with member services and administrative capabilities without charging an engineering fee.
- Enabled web-based member and administrative workflows by integrating React interfaces with a Node.js/Express API and PostgreSQL database, then deploying the complete application on the association's Linux VPS.

SELECTED SOFTWARE PROJECTS

ForgeRL / ForgeBench | Demo & evidence

- Built an isolated coding-agent evaluation framework with durable budget controls, completing a frozen prospective pilot of 162 episodes across 27 miniature tasks and six policies with full traces and a separate source-derived track. Limited generalization with one seed and six test families while keeping model weights unchanged.

MTA Scan | Live demo

- Built a Mapbox transit workspace with autonomous data collection and 15- and 30-minute evaluations of feed-predicted arrival-spacing proxies. Used chronological holdouts and baseline comparisons, limiting conclusions to early feasibility without a validated incident claim.

Inference Systems Lab | Code & results

- Ran 1,536 measured requests on an RTX 4090 with pinned vLLM/Qwen versions, concurrency sweeps and direct GPU telemetry. Recorded prefix-cache comparisons and output mismatches; no lossless speedup claim.

EDUCATION

Bachelor of Science, Computer Science **Jun 2026**

National and Kapodistrian University of Athens

- Department of Informatics and Telecommunications | Degree: Very Good

TECHNICAL SKILLS

- **ML**: Python, PyTorch, NumPy, SciPy, scikit-learn, generative models, model-based RL, uncertainty analysis and experimental evaluation
- **Systems**: FastAPI, Node.js/Express, React, JavaScript, SQL/PostgreSQL, Redis, Linux, Nginx, Docker, Git, automated testing

RESEARCH

Score-based MRI reconstruction

BSc thesis | Excellent (10/10) | Reproducible evidence

Reimplemented a published MRI reconstruction method in PyTorch with Flax checkpoint conversion. The thesis reports PSNR agreement within 0.15 dB on 256 BraTS slices at 4x, 8x and 24x undersampling. Released CPU evidence and ensemble-diagnostic tools; original prior credited, retrospective research.

Latent world-model learning and evaluation

Independent study · ongoing and unpublished

Investigate imagination horizons, critic stability and policy selection with PyTorch latent world models. Design controlled ablations and paired simulator evaluations, retaining task-dependent outcomes and failed hypotheses.